
Group13 – CDH

Employment website
Software development plan
Version 1.0

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Software development plan	Date: 18/10/2025

Revision History

Date	Version	Description	Author
16/10/2025	1.0	Initial version of project plan draft	Trần Nhật Dương
18/10/2025	1.1	Final version of project plan	Trần Nhật Dương
03/11/2025	1.2	Updation in phase plan	Đào Duy Hào, Nguyễn Đình Chuẩn

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Software Development Plan

1. Introduction

The introduction of the Software Development Plan provides an overview of the entire document. It includes the purpose, scope, definitions, acronyms, abbreviations, references, and overview of this Software Development Plan.

1.1 Purpose

The purpose of the *Project Plan* is to gather all information necessary to control the project. It describes the approach to the development of the software and is the top-level plan generated and used by managers to direct the development effort.

The following people use the *Software Development Plan*:

- The **project manager** uses it to plan the project schedule and resource needs, and to track progress against the schedule.
- **Project team members** use it to understand what they need to do, when they need to do it, and what other activities they are dependent upon.

1.2 Scope

This *Software Development Plan* describes the overall plan to be used by the CDH project. The plans as outlined in this document are based upon the product requirements as defined in the *Vision Document*.

1.3 Overview

This *Project Plan* contains the following information:

Project Overview — provides a description of the project's purpose, scope, and objectives. It also defines the deliverables that the project is expected to deliver.

Project Organization — describes the organizational structure of the project team.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

The main purpose of this project is to develop an efficient and user-friendly **Employment Website** that simplifies and digitalizes the recruitment process. It aims to create a convenient platform where job seekers can easily find and apply for suitable jobs, and employers can post vacancies, review applications, and manage recruitment activities effectively.

The objectives of the project are:

- To design and implement a complete **web application** integrating frontend, backend, and database layers.
- To ensure **secure and smooth user authentication** for different roles (applicants, employers, and admins).
- To provide **interactive and responsive interfaces** for both desktop and mobile browsers.
- To implement **core functionalities** such as job searching, application tracking, employer management, and admin supervision.
- To gain **hands-on experience** in applying modern web development technologies and collaborative teamwork practices.

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The expected deliverables include a fully functional Job Portal web application, a structured database for managing users and job data, and complete documentation with system design, testing results, and a final presentation demonstrating the system's main features.

2.2 Assumptions and Constraints

Assumptions:

- Some team members are not yet familiar with web development concepts but are willing to learn and improve during the project.
- The required development tools, frameworks, and libraries are open-source or freely available.
- Online learning resources and documentation are accessible throughout the project period.

Constraints:

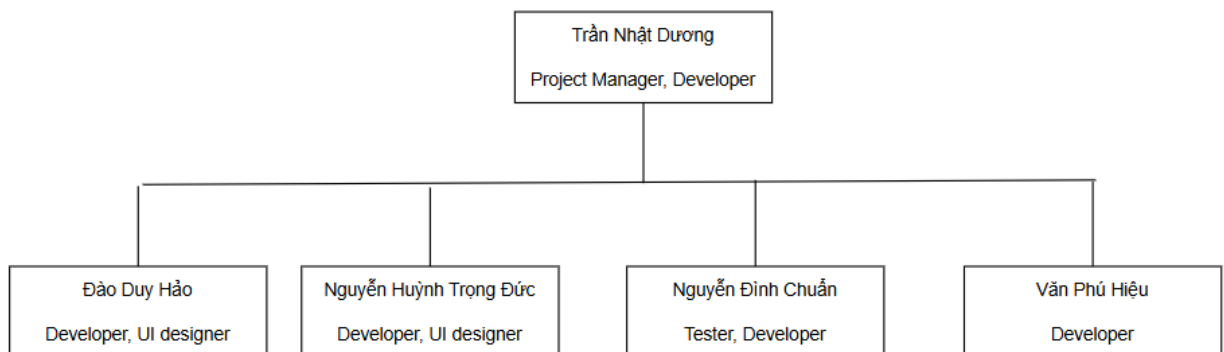
- The project must be completed within the semester's timeline
- There is no external budget for paid hosting or professional tools.
- The team size is small, which may affect task distribution and development speed.

2.3 Project Deliverables

Artifact	Target delivery dates
Vision document	24/10/2025
Project plan	24/10/2025
Use case model	6/11/2025
Use case specification	6/11/2025
Software architecture development	20/11/2025
Class diagram	20/11/2025

3. Project Organization

3.1 Organizational Structure



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3.2 Roles and Responsibilities

Person	Role
Trần Nhật Dương, Team Leader, Developer, Business Analyst	Responsible for overall project management, task distribution, server development and maintenance, as well as communication and feedback collection with TA.
Đào Duy Hào, Developer, UI designer	Responsible for job one type of user UI design and developing the frontend pages of the website.
Nguyễn Huỳnh Trọng Đức, Developer, UI designer	Designs and develops the user interface (UI) of another user to ensure a user-friendly and visually appealing experience.
Nguyễn Đình Chuẩn, Test manager, Developer	Assisting the project manager in recording each sprint, implementing the test plan for the website, developing the UI team in frontend and support team for writing meeting records in each sprint.
Văn Phú Hiệu, Developer	Responsible for developing the user interface, integrating APIs to fetch data, and handling logic processing with the server.

4. Management Process

4.1 Project Plan

4.1.1 Phase Plan

The Job Portal project is divided into three main phases following the Rational Unified Process (RUP): **Inception**, **Elaboration**, and **Construction**. The development process is organized into **six sprints**, each corresponding to one Project Assignment (PA), ensuring steady progress and clear deliverables throughout the project.

- **Inception Phase (Sprint 1-2 / PA0-PA1):** The team proposes the project idea, identifies target users, outlines main features, and develops a project plan that includes scope, timeline, and constraints. Furthermore, implement and outline basic UI design and web pages.
- **Elaboration Phase (Sprint 3-4-5 / PA2-PA4):** This phase focuses on developing the main structure of the website while finalizing essential documentation such as the requirement specification, use case model and software architecture. The team finalizes requirements, designs core system components (frontend, backend, database), and builds a solid foundation for implementation.
- **Construction Phase (Sprint 5-6 / PA5-PA6):** This phase involves implementation, testing, and final release. The team develops the full web application, performs testing to ensure performance and reliability, and completes the final release and prepare for the project defense with TA.

4.1.2 Schedule timeline

PA0 - Week 1 - 27/09/2025

Focus: Discuss and finalize the project topic, define target users and key features, and prepare the team contract and project proposal.

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Goal: All members of project team get familiar with required programming language, complete project idea with target users and key features

Task	Member	Timeline
Generate project idea with target users, key features	All members of team	29/9/2025
Complete team contract, project proposal content	All member of team	29/9/2025
Get familiar with required programming language	All member of team	29/9/2025
Prepare for team contract, project proposal	Trần Nhật Dương	29/9/2025

PA0 - Week 2 - 4/10/2025

Focus: Basic outline of code structure and UI design

Goal:

- Design and implement basic UI for login and register pages.
- Develop initial home pages for Employer and Candidate.
- Set up the database and API connection for basic data handling.
- Prepare project proposal and team contract documentation.

Task	Member	Deadline
Implement login page	Nguyễn Huỳnh Trọng Đức	13/10/2025
Implement register page	Văn Phú Hiệu	13/10/2025
Candidate home page design and develop	Đào Duy Hào	20/10/2025
Employer home page design and develop	Nguyễn Đình Chuẩn	20/10/2025
Basic database design and api setup	Trần Nhật Dương	20/10/2025
Prepare for team contract, project proposal	Trần Nhật Dương	11/10/2025

PA1 — Week 1 - 11/10/2025

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Focus: Basic Employer/Candidate frontend, backend outline

Goal: Build basic frontend pages for Employer and Candidate, and outline backend structure with initial database design and API connection.

Task	Member	Timeline
Merge code between login page and register page, API connection to server	Nguyễn Huỳnh Trọng Đức, Văn Phú Hiệu	20/10/2025
Candidate home page design and develop	Đào Duy Hào	20/10/2025
Employer home page design and develop	Nguyễn Đình Chuẩn	20/10/2025
Database design, API outline	Trần Nhật Dương	20/10/2025
Prepare documentation	All member of team	25/10/2025
Prepare report	Nguyễn Đình Chuẩn, Đào Duy Hào	25/10/2025

PA1 — Week 2 - 18/10/2025

Focus: Implement key job-related features for Employer and Candidate, and integrate updated API with backend.

Goal:

- Develop core job portal features including job posting, job listing with pagination, and CV upload.
- Enable Employers to manage job posts and candidate CVs effectively.
- Update and refine API endpoints to ensure smooth frontend-backend communication.
- Prepare project documentation for review.

Task	Member	Timeline
Job posting form of employer UI	Nguyễn Đình Chuẩn	27/10/2025
CV management of employer	Nguyễn Huỳnh Trọng Đức	27/10/2025
Job listing with pagination	Đào Duy Hào	27/10/2025
Job filtering + CV upload	Văn Phú Hiệu	27/10/2025
Update API endpoint	Trần Nhật Dương	27/10/2025

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Prepare documentation	All member of team	25/10/2025
Prepare report	Nguyễn Đình Chuẩn, Đào Duy Hào	25/10/2025

PA2 - Week 1 - 25/10/2025

Focus: Finalize profile management and job post functions, and complete backend API integration.

Goal:

- Implement display and update features for both Employer and Candidate profiles.
- Develop and connect job post update functions with the backend API.
- Handle API processing for retrieving job lists and job details.
- Finalize API endpoints to ensure full communication between frontend and backend.
- Complete documentation for current system development progress.

Task	Member	Timeline
Employer profile (display, update)	Nguyễn Đình Chuẩn	3/11/2025
Candidate profile (display, update)	Đào Duy Hào	3/11/2025
Job post update	Nguyễn Huỳnh Trọng Đức	3/11/2025
API processing for getting job list and job detail	Văn Phú Hiệu	3/11/2025
Complete API endpoint	Trần Nhật Dương	3/11/2025
Prepare use-case model	Văn Phú Hiệu	8/11/2025
Prepare employer use-case specification	Đào Duy Hào, Nguyễn Đình Chuẩn	8/11/2025
Prepare candidate use-case specification	Trần Nhật Dương, Nguyễn Huỳnh Trọng Đức	8/11/2025
Prepare report	Trần Nhật Dương, Văn Phú Hiệu	25/10/2025

PA2 - Week 2 - 1/11/2025

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Focus: Complete key features include: response to candidate, CV label of employers, tracking application status

Goal:

- Implement the **response-to-candidate** feature, allowing employers to mark applications as viewed, rejected, or accepted.
- Develop **CV labeling** for employers to organize and manage received CVs effectively.
- Build the **application tracking** system for candidates to monitor employer responses and job application progress.
- Add the **job saving** function to let candidates save preferred job postings.
- Integrate **backend authorization** and **admin login** to secure system access and ensure role-based authentication.

Task	Member	Timeline
Response to candidate (reject, viewed, ...)	Nguyễn Huỳnh Trọng Đức	10/11/2025
CV labeling	Nguyễn Đình Chuẩn	10/11/2025
Tracking application, receive response from employer	Văn Phú Hiệu	10/11/2025
Job saving	Đào Duy Hào	10/11/2025
Authorization for backend	Trần Nhật Dương	10/11/2025
Admin login	Trần Nhật Dương	10/11/2025
Prepare use-case model	Văn Phú Hiệu	8/11/2025
Prepare employer use-case specification	Đào Duy Hào, Nguyễn Đình Chuẩn	8/11/2025
Prepare candidate use-case specification	Trần Nhật Dương, Nguyễn Huỳnh Trọng Đức	8/11/2025
Update vision document, project plan	Văn Phú Hiệu, Trần Nhật Dương	8/11/2025
Prepare report	Văn Phú Hiệu, Trần Nhật Dương	8/11/2025

PA3 - Week 1 - 8/11/2025

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Focus: Enhancement of the system's User Interface and User Experience (UI/UX).

Goal: To improve the platform's usability, visual design

Task	Member	Timeline
UI update	All member of team	17/11/2025
Admin login	Trần Nhật Dương	17/11/2025
Prepare documentation	All member of team	21/11/2025
Prepare report	Đào Duy Hào, Nguyễn Huỳnh Trọng Đức	21/11/2025

PA3 - Week 2 - 15/11/2025

Focus: Implement core backend and frontend functionalities to enhance system reliability, automation, and user experience for both employers and job seekers.

Goal:

- Develop and integrate the **payment feature** to allow employers to post jobs securely.
- Implement **AI-based CV recommendation** to improve matching efficiency for employers.
- Add a **reset password function** to enhance account security and user accessibility.
- Build the **admin statistic module** to track system usage and recruitment activities.
- Finalize and submit **project documentation** summarizing system design and current progress.

Task	Member	
Payment for job posting	Nguyễn Đình Chuẩn	24/11/2025
AI CV recommendation for employers	Nguyễn Huỳnh Trọng Đức	24/11/2025
Reset password function	Văn Phú Hiệu, Đào Duy Hào	24/11/2025
Admin statistic function	Trần Nhật Dương	24/11/2025
Update use-case model, use-case specification	Đào Duy Hào	21/11/2025
Prepare software architecture	All member of team	21/11/2025
Prepare class diagram	Nguyễn Đình Chuẩn, Nguyễn Huỳnh Trọng Đức	21/11/2025

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Prepare report	Đào Duy Hào, Nguyễn Huỳnh Trọng Đức	21/11/2025
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PA4 - 22/11/2025

Focus: Initial release of Job portal

Goal: Bug process, UI updation

Task	Member	Timeline
Bug identification	Nguyễn Đình Chuẩn	12/12/2025
Bug fixing	All member of team	12/12/2025
UI/UX improvement	Đào Duy Hào, Nguyễn Huỳnh Trọng Đức	12/12/2025
Prepare for SAD revision	All member of team	10/12/2025
Prepare report	Đào Duy Hào, Nguyễn Huỳnh Trọng Đức	10/12/2025

PA5 - 15/12/2025

Focus: The second release of Job portal

Goal: To deliver the final, fully functional version of the Job Portal with complete features, improved performance, and enhanced user experience.

Task	Member	Timeline
Bug identification	Nguyễn Đình Chuẩn	20/12/2025
Bug fixing	All member of team	20/12/2025
UI/UX improvement	Đào Duy Hào, Nguyễn Huỳnh Trọng Đức	20/12/2025
Prepare documentation	All members of team	18/12/2025
Prepare report	Nguyễn Đình Chuẩn, Đào Duy Hào	18/12/2025

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4.2 Project Monitoring and Control

4.2.1 Requirements Management

The requirements for this system are captured in the Vision document. Requested changes to requirements are captured in Change Requests, and are approved as part of the Configuration Management process.

4.2.2 Reporting and Measurement

Updated cost and schedule estimates, and metrics summary reports, will be generated at the end of each iteration.

The Minimal Set of Metrics, as described in the RUP [Guidelines: Metrics](#), will be gathered on a weekly basis. These include:

Earned value for completed tasks. This is used to re-estimate the schedule and budget for the remainder of the project, and/or to identify need for scope changes.

Total defects open and closed – shown as a trend graph. This is used to help estimate the effort remaining to correct defects.

Acceptance test cases passing – shown as a trend graph. This is used to demonstrate progress to stakeholders.

In addition, overall costs will be monitored against the project budget.

4.2.3 Risk Management

Risks will be identified in the Inception Phase using the steps identified in the RUP for Small Projects activity “Identify and Assess Risks”. Project risk is evaluated at least once per iteration and documented in this table. The risks of the greatest magnitude are listed first in the table.

Risk Ranking (High, Medium, Low)	Risk Description and Impact	Mitigation Strategy and/or Contingency Plan
Low	Team member get sick	Delay the deadline of that member or distribute another member for support
Medium	Imbalance in work division among team members.	Ensure tasks are distributed evenly based on members
High	A team member may give up or withdraw from the project for personal or other reasons.	Maintain regular communication to identify potential issues early; ensure tasks are documented and shared; motivate and support team morale
High	Not consistency in coding convention	Defined common coding convention in the inception phase of project
Medium	Team members are not familiar with required language of project	Conduct short training sessions or self-learning plans before implementation

4.2.4 Configuration Management

Appropriate tools will be selected which provide a database of Change Requests and a controlled versioned repository of project artifacts.

All source code, test scripts, and data files are included in baselines. Documentation related to the source code is also included in the baseline, such as design documentation. All customer deliverable artifacts are included in the final baseline of the iteration, including executables.